

→ Linear Discriminant Analysis (LDA) is also k/a normal discriminant analysis (NDA) or ~~discriminant~~ discriminant function analysis (DFA)

→ LDA works by identifying a linear combination of features that separates two or more classes of object or events.

↳ LDA does this by projecting data with two or more ~~classes~~ of object dimension into one dimension so that it can be more easily classified.

↳ Therefore, sometime referred to as dimension reduction.

↳ LDA can be used for multi-class data classification problem.

↳ LDA is thus often applied to enhance the operation of other learning classification algorithms such as decision tree, random forest or support vector ~~mach~~ machine (SVM)

↳ It supervised machine learning algo.

↳ It is used to project the features in higher dimensional space into a lower dimension space.

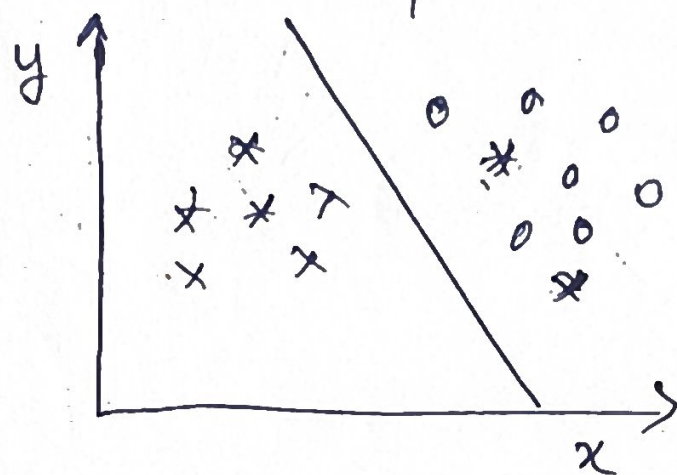
↳ LDA assumes that the data has Gaussian distribution and that the covariance matrices of different classes are equal.

↳ It also assumes that the data is linearly separable. ①

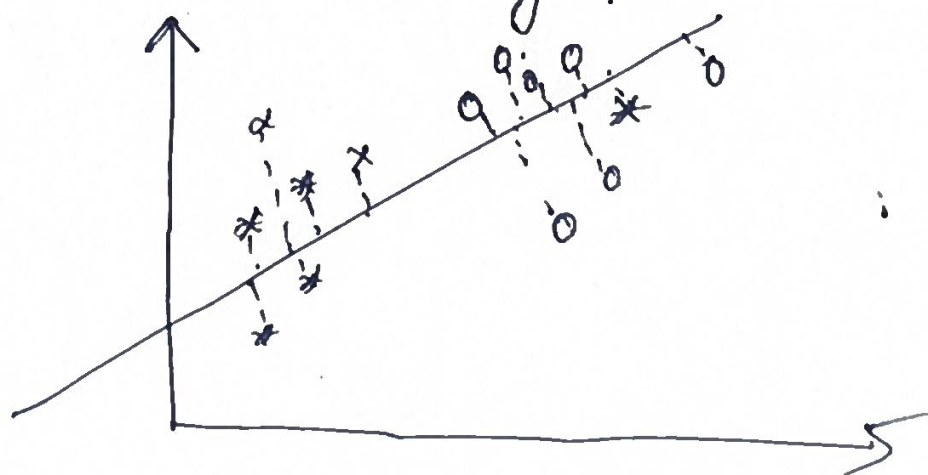
Qx

Suppose we have two sets of data points belonging to two different classes that we want to classify.

• When the data points are plotted on 2D plane, there's no straight line that can separate the two classes of the data points completely.



• Here, Linear Discriminant Analysis (LDA) uses both the axes (x and y) to create a new axis and projects data onto a new axis in a way to maximize the separation of the two categories and hence, reduce the 2D graph into a 1D graph.



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• Two criteria are used by LDA to create a new axis:

↳ Maximize the distance b/w means of two classes.

↳ Minimize the variation within each class.



New axis

Steps for convert higher dimension to lower dimension for LDA

1. Compute the class means of dependent variable $\mu_i = \frac{1}{N} \sum_{x \in \omega_i} x$

2. Derive the covariance matrix of class variable

$$S_i = \sum_{x \in \omega_i} (x - \mu_i)(x - \mu_i)^T$$

similarly do for 2 class, 3 class so on.

3. Compute the within class scatter matrix

$$S_w = S_1 + S_2$$

4. Compute the between class matrix scatter

$$S_B = (\mu_1 - \mu_2)(\mu_1 - \mu_2)^T$$

- ④ 5. Compute the eigen values and eigen vectors from the within class and between class scatter matrix

$$S_W^{-1} S_B W = \lambda W$$

6. Sort the value of eigen values and select the top K values.

7. Find the eigen vectors corresponds to the top K eigen vectors.

$$[S_W^{-1} S_B - \lambda I] \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = 0$$

8. Obtain the LDA by taking the dot product of eigen vectors and original data.

Ex Compute the Linear Discriminant (5)
projection for the following two dimensional
dataset.

sample for class ω_1 : $X_1 = (x_1, x_2) = \{(1, 2), (2, 1), (2, 3), (3, 6), (4, 4)\}$

sample for class ω_2 : $X_2 = (x_1, x_2) = \{(5, 10), (6, 8), (9, 5), (8, 7), (10, 2)\}$

The classes mean are.

$$\mu_1 = \frac{1}{N_1} \sum_{x \in \omega_1} x = \frac{1}{5} \left[\begin{pmatrix} 1 \\ 2 \end{pmatrix} + \begin{pmatrix} 2 \\ 1 \end{pmatrix} + \begin{pmatrix} 2 \\ 3 \end{pmatrix} + \begin{pmatrix} 3 \\ 6 \end{pmatrix} + \begin{pmatrix} 4 \\ 4 \end{pmatrix} \right] = \begin{pmatrix} 3 \\ 3.8 \end{pmatrix}$$

$$\mu_2 = \frac{1}{N_2} \sum_{x \in \omega_2} x = \frac{1}{5} \left[\begin{pmatrix} 5 \\ 10 \end{pmatrix} + \begin{pmatrix} 6 \\ 8 \end{pmatrix} + \begin{pmatrix} 9 \\ 5 \end{pmatrix} + \begin{pmatrix} 8 \\ 7 \end{pmatrix} + \begin{pmatrix} 10 \\ 2 \end{pmatrix} \right] = \begin{pmatrix} 8.4 \\ 7.6 \end{pmatrix}$$

$$\begin{aligned} S_1 &= \sum \frac{(x - \mu_1)(x - \mu_1)^T}{N-1} = \frac{1}{4} \left[\begin{pmatrix} 1 \\ 2 \end{pmatrix} - \begin{pmatrix} 3 \\ 3.8 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} - \begin{pmatrix} 3 \\ 3.8 \end{pmatrix}^T \right. \\ &\quad + \begin{pmatrix} 2 \\ 1 \end{pmatrix} - \begin{pmatrix} 3 \\ 3.8 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} - \begin{pmatrix} 3 \\ 3.8 \end{pmatrix}^T + \begin{pmatrix} 2 \\ 3 \end{pmatrix} - \begin{pmatrix} 3 \\ 3.8 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} - \begin{pmatrix} 3 \\ 3.8 \end{pmatrix}^T \\ &\quad + \begin{pmatrix} 3 \\ 6 \end{pmatrix} - \begin{pmatrix} 3 \\ 3.8 \end{pmatrix} \begin{pmatrix} 3 \\ 6 \end{pmatrix} - \begin{pmatrix} 3 \\ 3.8 \end{pmatrix}^T + \begin{pmatrix} 4 \\ 4 \end{pmatrix} - \begin{pmatrix} 3 \\ 3.8 \end{pmatrix} \begin{pmatrix} 4 \\ 4 \end{pmatrix} - \begin{pmatrix} 3 \\ 3.8 \end{pmatrix}^T \Big] \\ &= \begin{pmatrix} 1 & -0.25 \\ -0.25 & 2.2 \end{pmatrix} \end{aligned}$$

$$(6) S_2 = \sum_{x \in \omega_2} \frac{(x - \mu_2)(x - \mu_2)^T}{N-1}$$

$$= \left[\begin{pmatrix} 9 \\ 10 \end{pmatrix} - \begin{pmatrix} 8.4 \\ 7.6 \end{pmatrix} \right] \left[\begin{pmatrix} 9 \\ 10 \end{pmatrix} - \begin{pmatrix} 8.4 \\ 7.6 \end{pmatrix} \right]^T + \left[\begin{pmatrix} 6 \\ 8 \end{pmatrix} - \begin{pmatrix} 8.4 \\ 7.6 \end{pmatrix} \right] \left[\begin{pmatrix} 6 \\ 8 \end{pmatrix} - \begin{pmatrix} 8.4 \\ 7.6 \end{pmatrix} \right]^T$$

$$+ \left[\begin{pmatrix} 9 \\ 5 \end{pmatrix} - \begin{pmatrix} 8.4 \\ 7.6 \end{pmatrix} \right] \left[\begin{pmatrix} 9 \\ 5 \end{pmatrix} - \begin{pmatrix} 8.4 \\ 7.6 \end{pmatrix} \right]^T + \left[\begin{pmatrix} 8 \\ 7 \end{pmatrix} - \begin{pmatrix} 8.4 \\ 7.6 \end{pmatrix} \right] \left[\begin{pmatrix} 8 \\ 7 \end{pmatrix} - \begin{pmatrix} 8.4 \\ 7.6 \end{pmatrix} \right]^T$$

$$+ \left[\begin{pmatrix} 10 \\ 8 \end{pmatrix} - \begin{pmatrix} 8.4 \\ 7.6 \end{pmatrix} \right] \left[\begin{pmatrix} 10 \\ 8 \end{pmatrix} - \begin{pmatrix} 8.4 \\ 7.6 \end{pmatrix} \right]^T \bigg/ N-1$$

$$= \begin{pmatrix} 2.3 & -0.05 \\ -0.05 & 3.3 \end{pmatrix}$$

• Within-class scatter matrix

$$S_W = S_1 + S_2$$

$$= \begin{pmatrix} 1.8 & -0.25 \\ -0.25 & 2.2 \end{pmatrix} + \begin{pmatrix} 2.3 & -0.05 \\ -0.05 & 3.3 \end{pmatrix}$$

$$= \begin{pmatrix} 3.3 & -0.3 \\ -0.3 & 5.5 \end{pmatrix}$$

• Between class scatter matrix

$$S_B = (\mu_1 - \mu_2)(\mu_1 - \mu_2)^T$$

$$= \left[\begin{pmatrix} 3 \\ 3.8 \end{pmatrix} - \begin{pmatrix} 8.4 \\ 7.6 \end{pmatrix} \right] \left[\begin{pmatrix} 3 \\ 3.8 \end{pmatrix} - \begin{pmatrix} 8.4 \\ 7.6 \end{pmatrix} \right]^T$$

$$= \begin{pmatrix} -5.4 \\ -3.8 \end{pmatrix} \begin{pmatrix} -5.4 & -3.8 \end{pmatrix}$$

$$= \begin{pmatrix} 29.16 & 20.52 \\ 20.52 & 14.44 \end{pmatrix}$$

Find Eigen value.

(7)

$$S\omega^{-1}S_B\omega = \lambda\omega \Rightarrow |S\omega^{-1}S_B - \lambda I| = 0$$

$$= \left| \begin{pmatrix} 3.3 & -0.3 \\ -0.3 & 5.5 \end{pmatrix}^{-1} \begin{pmatrix} 29.16 & 20.52 \\ 20.52 & 14.41 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right| = 0$$

$$= \left| \begin{pmatrix} 0.3045 & 0.0166 \\ 0.0166 & 0.1827 \end{pmatrix} \begin{pmatrix} 29.16 & 20.52 \\ 20.52 & 14.41 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right| = 0$$

$$= \left| \begin{pmatrix} 9.2213 - \lambda & 6.489 \\ 4.2339 & 2.9794 - \lambda \end{pmatrix} \right|$$

$$= (9.2213 - \lambda)(2.9794 - \lambda) - 6.489 \times 4.2339 = 0$$

$$= \lambda^2 - 12.2007\lambda = 0$$

$$= \lambda(\lambda - 12.2007) = 0$$

$$= \lambda_1 = 0, \lambda_2 = 12.2007$$

• Find eigen vector

$$(S\omega^{-1}S_B - \lambda I) \begin{pmatrix} \omega_1 \\ \omega_2 \end{pmatrix} = 0$$

$$\omega_1 = \begin{pmatrix} -0.5755 \\ 0.8178 \end{pmatrix}$$

$$\omega_2 = \begin{pmatrix} 0.9688 \\ 6.4173 \end{pmatrix} = \omega^*$$

obtain the LDA by taking the dot product of eigen vectors and original data (9)

$$w_2 = \begin{pmatrix} 0.9088 \\ 0.4173 \end{pmatrix} \begin{matrix} 1 \\ 2 \end{matrix}$$

~~samples~~ f

x_1	4	2	2	3	4	9	6	9	8	10
x_2	2	4	3	6	4	10	8	5	7	8
LD	4.46	3.78	3.06	5.2	5.3	12.35	8.8	10.2	10.19	12.42

3.635
-9742